TEXORA · COURSE NOTES

Introduction to Textile

What a textile is, how the craft grew over thousands of years, and how raw fibre becomes a finished product.

TE-1109 · FUNDAMENTALS OF TEXTILE ENGINEERING · TOPIC 1 · LEVEL 1 TERM 1

1. What is a textile?

The word *textile* comes from the Latin *texere*, meaning “to weave”, and from *textilis*, “woven”. In its oldest and narrowest sense, a textile is simply a cloth made by weaving — yarns crossing each other at right angles on a loom. That definition served well for centuries, because weaving was the main way of turning fibre into fabric.

Today the word covers far more. Knitting, braiding, felting, bonding and several other methods also turn fibres into sheet-like materials, and none of them is weaving. The modern definition is therefore broader:

> **DEFINITION**
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> A textile is a flexible material made from fibres or yarns — by weaving, knitting, braiding, felting, bonding or similar processes — used for clothing, household goods, and a wide range of technical and industrial purposes.

Three ideas inside this definition matter. First, the starting point is always a **fibre** — a thin, hair-like unit whose length is many times its diameter. Second, fibres are usually (not always) converted into **yarn** before they become fabric. Third, the end product is **flexible**: drape and softness are what separate textiles from rigid materials such as plastic sheet or metal foil, even though modern technical textiles can be surprisingly strong.

Textiles are also one of the basic needs of human life. Along with food and shelter, clothing protects the body from heat, cold, sun and injury — and, over time, it became a medium of comfort, identity and fashion as well.

2. A short chronological development

Textile making is older than writing. The main stages of its development are summarised below.

PERIOD	DEVELOPMENT

Prehistoric era	Early humans use animal skins and plant fibres for covering. Flax is cultivated and spun into linen in Egypt and Mesopotamia; cotton is grown and woven in the Indus Valley; silk is produced in China.
Ancient world	The hand spindle and simple looms spread widely. Fine cotton muslin from Bengal becomes famous across the Roman world.
Middle ages	Wool dominates Europe; guilds organise spinning and weaving as skilled crafts. The spinning wheel reaches Europe from Asia and raises yarn output.
1733 – 1785	The Industrial Revolution begins in textiles: the flying shuttle (1733), spinning jenny (1764), water frame (1769), spinning mule (1779) and the power loom (1785) move production from homes into factories.
19th century	Steam power, the sewing machine and chemical dyes transform output and cost. Fibre production and trade become global.
20th century	Man-made fibres arrive — rayon, then nylon (1938), polyester and acrylic — alongside high-speed machinery and synthetic dyes.
Today	Automation, computer-aided design and technical textiles: fabrics that filter, conduct, protect, heal or sense, far beyond clothing alone.

Notice the pattern: for thousands of years the *methods* barely changed — only skill and scale grew. Then, in about fifty years, machinery changed everything, and textiles became the first truly industrialised sector. This is why the textile industry is often called the mother of industrialisation.

3. From fibre to finished product: the process flow

Every textile product, whether a handloom sari or a factory-made shirt, passes through the same basic chain of stages. Understanding this chain is the foundation of the whole subject — almost every later course in this degree studies one link of it in depth.

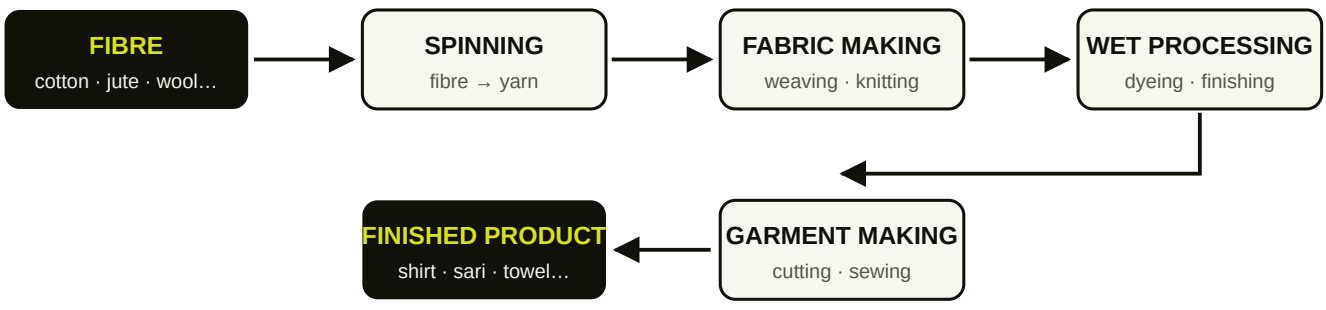


Figure 1 — The basic process flow of textile materials, from raw fibre to finished product.

Fibre. The chain begins with raw fibres — natural ones such as cotton, jute, wool and silk, or man-made ones such as polyester and nylon. Fibres are cleaned, opened and aligned.

Spinning. Fibres are drawn out and twisted together into a continuous strand called yarn. Twist gives the yarn its strength.

Fabric making. Yarns are interlaced by weaving, interlooped by knitting, or formed into fabric by nonwoven methods. At this stage the material is called grey (or greige) fabric — unfinished and usually off-white.

Wet processing. The grey fabric is prepared (desizing, scouring, bleaching), coloured (dyeing or printing) and finished (softening, mercerising and other treatments) to give it the look, feel and performance the buyer wants.

Garment making and finishing. Finally the fabric is cut into pattern pieces, sewn into garments or other products, pressed, checked and packed — the finished product ready for the user.

> **KEY POINT**
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> Each stage adds value and cost. A fault introduced at an early stage — poor fibre, weak yarn — cannot be repaired later; it can only be carried, expensively, to the end. This is why quality control in textiles starts at the fibre, not at the final inspection.

4. Summary

- A textile is a flexible material built from fibres or yarns; the word comes from the Latin *texere*, “to weave”.
- The craft is ancient, but its transformation came with 18th-century machinery — textiles led the Industrial Revolution.
- Production follows a fixed chain: **fibre → spinning → fabric making → wet processing → garment making → finished product**.
- Almost every course in textile engineering studies one stage of this chain in depth.

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